

NEXT-2 â Paperclip Sandbox Service Recovery

Objective

Move Paperclip from 'paperclip_sandbox_service_not_running' toward live health proof in local/Tailnet-safe mode.

Result

PARTIAL+ (Tailnet Paperclip health is reachable directly, and Mission Control now includes an automatic Tailnet fallback candidate derived from the Mission Control public host; owner login proof is still pending).

Actions Executed

1. Probed local loopback Paperclip health endpoint.
2. Probed Tailnet Paperclip health endpoint.
3. Implemented Paperclip endpoint candidate fallback in runtime code:
 - when loopback is unavailable and no explicit Paperclip endpoint is set,
 - derive Tailnet host from 'NEXT_PUBLIC_APP_URL' / 'MC_PUBLIC_BASE_URL' / 'APP_URL' / 'MISSION_CONTROL_PUBLIC_URL',
 - probe 'http(s)://<tailnet-host>:3100'.
4. Added regression test covering loopback failure + Tailnet fallback success.
5. Confirmed no public exposure changes were introduced.
6. Prepared exact owner/admin runtime wiring package for cases where environment requires explicit endpoint pinning.

Proof

- 'http://127.0.0.1:3100/api/health' -> unreachable from this runtime context.
- 'http://100.116.35.95:3100/api/health' -> '200' with '{"status":"ok","deploymentMode":"authenticated","bootstrapStatus":"ready",...}'.

Interpretation

- Paperclip service is reachable on Tailnet endpoint.
- Bridge runtime now has a safe automatic Tailnet fallback path for Paperclip health checks.
- Paperclip remains non-GO because owner login/session and live workforce data routes are still not proven.

Exact Owner/Admin Action Package

1. Set Mission Control runtime Paperclip endpoint via service manager (not '.env' file edits):
 - 'PAPERCLIP_API_URL=http://100.116.35.95:3100'
2. Restart Mission Control runtime.
3. Re-run:
 - 'GET /api/bridge/paperclip/status'
 - 'GET /api/bridge/paperclip/companies'
 - 'GET /api/bridge/paperclip/agents'
 - 'GET /api/bridge/paperclip/issues'
4. Keep writes blocked until Bridge Session scope and adapter proof are complete.

Current Blockers

- 'paperclip_owner_session_required'
- 'paperclip_auth_required_or_not_configured' (for company/agent/issue reads when owner session is absent)

Safety Confirmations

- No public exposure added.
- No production secrets used.
- No '.env' changes.
- No auth weakening.

Files Changed

- 'src/lib/paperclip-bridge.ts'
- 'src/lib/paperclip-bridge.test.ts'
- 'runtime/next-paperclip-sandbox-service-recovery.md'
- 'runtime/next-paperclip-sandbox-service-recovery.pdf'

Next Step

After endpoint/runtime wiring, re-run Paperclip bridge read-only routes and owner-login lane.